



Data-Based Modeling of Power-Split Hybrids Using Cascaded Neural Networks

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Abstract

Power-split hybrid powertrains represent one of the most advanced and complex types of powertrain systems. The combination of multiple energy sources and power paths offers great potential but results in complex interactions that require improved strategies for optimal efficiency and emission control. The development and optimization of such operating strategies typically involve algorithms that demand fast computational environments. Traditional high-accuracy numerical simulations of such a complex system are computationally expensive, limiting their applicability for extensive iterative optimizations and real-time applications. This paper introduces a data-based approach designed specifically to address this challenge by efficiently modeling the dynamic behavior of power-split hybrid powertrains using cascaded neural networks.

Cascaded neural networks consist of interconnected subnetworks, each specifically trained to represent individual drivetrain components or subsystems. This modular

structure allows the networks to cover a larger parameter space within each subnetwork effectively and enables the generation of larger and more targeted training datasets using simpler, separated numerical models. However, this approach is also challenging due to the need for step prediction, potential error accumulation and the requirement for feedback loops. Assumptions are necessary to overcome these limitations and improve the model predictions. The focus is on the possible connection to optimization algorithms with targeting key parameters such as battery state of charge, vehicle velocity, and emissions profiles.

By significantly reducing computational demands compared to complex and detailed simulations, the cascaded data-driven models state a promising basis for real-time applications, enabling sophisticated control strategies tailored to the complexities inherent to hybrid electric vehicles. This method accelerates the optimization process, enhances adaptability and scalability of control strategies, and significantly contributes to the development of cleaner, more efficient vehicles.

Introduction

The automotive sector is characterized by rapid development and constant technological evolution. Driven by increasing competition, tightening regulatory requirements – especially regarding emissions – and the push towards electrification, modern powertrains are becoming increasingly complex. A prime example of this trend is the power-split hybrid powertrain, which combines multiple energy sources and mechanical paths in a highly integrated layout [1]. Such complexity introduces challenges that may not be fully captured by traditional rule-based control strategies, despite their reliability and broad applicability. There remains significant potential for optimization – particularly in achieving efficiency and emissions targets across a wide range of operating conditions.

This is where neural network-based optimization algorithms offer a promising alternative. These algorithms rely on data-driven learning mechanisms, such as trial-and-error search strategies, to optimize toward defined objectives. However, their effectiveness depends critically on fast-running and highly efficient simulation environments – so-called digital twins [2].

Conventional numerical models, such as 0D/1D simulation tools, offer high accuracy but are often too computationally intensive for real-time applications or iterative optimization loops [3]. This limitation becomes especially problematic under transient conditions or non-standard configurations (e.g., pre-turbine aftertreatment). A promising alternative is the use of data-driven models, commonly implemented as artificial neural networks

(ANN). When trained on high-quality data – either from measurements or detailed simulations – neural networks, particularly those based on Long Short-Term Memory (LSTM) cells, can learn complex nonlinear system dynamics [4]. Their greatest strength lies in their extremely low inference time, enabling real-time or faster-than-real-time predictions, which makes their use for optimization algorithms more attractive and also enable them to be used on control units.

To further improve scalability and flexibility, this work explores the concept of cascaded neural networks, where individual subnetworks are trained to represent physically distinct subsystems. This modular approach offers advantages in data generation, training specialization, and system-wide coverage of the parameter space. It also allows for targeted refinement and easier model maintenance.

The objective of this work is to develop and evaluate two cascaded neural network approaches for replacing the physics-based simulation of a power-split hybrid powertrain. The system is divided into an internal combustion engine (ICE) model and a PG-EM model that includes the planetary gear, electric machines and the vehicle. While this modularization introduces several challenges – including prediction instabilities and error propagation – it also provides a framework for more flexible model design. A second cascaded model was developed to mitigate these issues using specific assumptions about the underlying control system. Both approaches are validated using the WLTC (Worldwide Harmonized Light Vehicles Test Cycle) driving cycle, assessing their ability to handle realistic transient conditions and identifying key areas for further refinement.

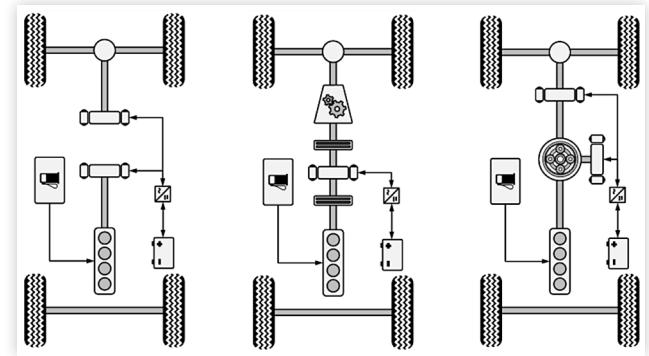
Theoretical Background

Hybrid Powertrain Topologies

The classification of hybrid systems is quite complex and typically based on the degree of hybridization, for which a comprehensive overview can be found in [1]. On the other hand, hybrid powertrain topologies are categorized according to the structural arrangement and interaction of energy converters and gearboxes. Typically, a hybrid powertrain integrates a conventional internal combustion engine (ICE) with one or more electric motors (EM) and employs a battery as the electric energy storage system. The primary hybrid powertrain topologies include parallel hybrids, series hybrids, and power-split hybrids [1]. A representation of the three configurations can be seen in Figure 1.

Power-Split Hybrid Power-split hybrids represent the most advanced and complex hybrid configuration, integrating features of both parallel and series hybrids. The core component of this system is a planetary gearbox, which connects the ICE, two electric motors, and the

FIGURE 1 Presentation of series- (left), parallel- (middle), and power-split (right) hybrid topology from [5]



driveshaft. This arrangement allows splitting the ICE-generated power into two paths: a mechanical path directly driving the vehicle and an electrical path producing energy via a generator to charge the battery. With a continuously variable transmission enabled by the planetary gearbox, the ICE speed is independent of vehicle speed, allowing operation at optimal efficiency [1]. A typical setup is the input-split configuration using one planetary gear set. Here, the ICE is connected to the carrier gear, one electric motor (EM1) to the sun gear, and the driveshaft and second electric motor (EM2) to the ring gear. Mechanical power from the ICE directly drives the ring gear via the carrier and indirectly drives the EM1 generator via the sun gear. The planetary gearbox allows ICE torque transmission to the ring gear only when EM1 exerts a negative torque on the sun gear, making EM1 primarily function as a generator [6]. EM1 thus controls the gear ratio, enabling continuously variable transmission, and produces electrical energy simultaneously. The EM2 on the ring gear can provide additional torque or independently drive the vehicle electrically, depending on configuration. Both electric motors can regenerate energy during braking events. Compared to other hybrid configurations, the power-split concept achieves an optimal balance between efficiency and performance and can adapt to various driving conditions such as urban and highway driving. Its complexity requires sophisticated control strategies. However, it particularly outplays its advantages in urban environments compared to optimized parallel hybrids. The EM2 is typically larger, enhancing regenerative braking capabilities, beneficial on highways. Although available in simpler single-stage planetary setups, more complex multi-stage configurations offer even finer adaptability at the cost of higher system complexity, increased control challenges, and greater powertrain losses. Due to its intricate design and higher component count, the power-split concept incurs higher manufacturing costs and vehicle weight [6]. Precise interaction between all components is essential for smooth vehicle operation. Particularly at high speeds, EM1 can operate inefficiently due to high rotational speeds in single-stage setups, whereas multi-stage designs require additional clutches and more complex controls. This work

specifically focuses on the single-stage planetary gear configuration.

Basics of Planetary Gears

The planetary gear plays a central role in power-split hybrid architectures and is key to enabling many of their functional advantages. To understand how the internal combustion engine's output can be split into mechanical and electrical power paths, this section introduces the basic principles of planetary gear systems [1].

A planetary gear set typically consists of three main components: a sun gear, planet gears mounted on a carrier, and a ring gear. This can be seen in Figure 2. The planet gears simultaneously mesh with the sun and ring gears and transfer torque between them. The system involves three rotating shafts: the sun gear (rotating at speed n_s), the carrier (n_c), and the ring gear (n_r). These three elements form a kinematic system with coupled rotational speeds. A key characteristic of a planetary gear system is the fixed gear ratio (i_{fix}) (1), defined when the carrier is held stationary.

$$i_{fix} = -\frac{z_r}{z_s} \quad (1)$$

while z_r describes the ring gear number of teeth and z_s the sun gear number of teeth. This ratio is equivalent with the relationship between the rotational speeds of the sun and ring gears. Due to the opposing rotational directions of the sun and ring gears in this scenario, the fixed gear ratio is negative [1].

The overall kinematic behavior of the planetary gear is governed by the Willis equation (2), which mathematically relates the rotational speeds of all three components, resulting in an equation with two degrees of freedom. Two of the speeds can be independently defined, while the third must be computed accordingly.

$$n_s - (1 - i_{fix})n_c - i_{fix}n_r = 0 \quad (2)$$

In contrast to speed relations, torque distribution in the planetary gear is more constrained. Only one of the

three shaft torques can be independently specified, while the others are determined by the internal structure, following equations (3) and (4). This implies that the torque on the carrier is inherently split between the sun and ring gears in a fixed proportion. These equations assume static behavior and neglect inertias [1].

$$\tau_s = -\frac{z_s}{z_s + z_r} \tau_c \quad (3)$$

$$\tau_r = -\frac{z_r}{z_s + z_r} \tau_c \quad (4)$$

$$\tau_c + \tau_s + \tau_r = 0 \quad (5)$$

The simple balance of torque in the system is described in equation (5). These characteristics are essential for understanding the energy flow and control in power-split hybrid drivetrains.

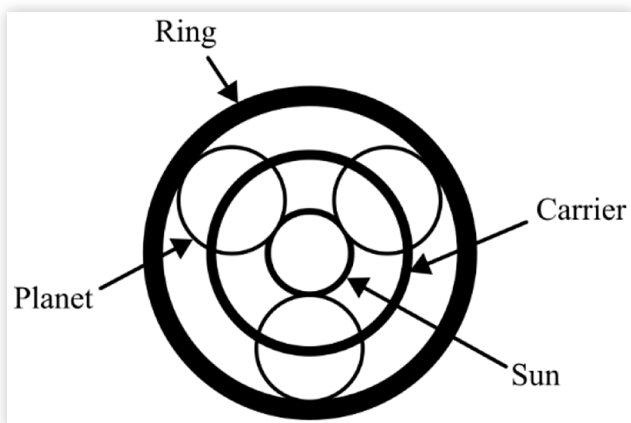
Operating Strategy in Power-Split Hybrids

An operating strategy defines the **temporal and logical sequence** of states for energy converters (ICE, electric motors), storage systems (battery), and possibly characteristic modifiers (e.g., motor map shaping). In hybrid vehicles, unlike conventional ICEs, **drive power can be split** between components to improve efficiency and reduce emissions [7].

Operating Modes Similar to other hybrid topologies, power-split hybrids offer several operational modes. One mode is **electric-only driving**, where EM2 exclusively powers the vehicle, and the ICE remains off, necessitating EM1 to rotate backward. EM1's speed limitation defines the vehicle's maximum electric-only speed. Another mode, known as **sailing**, occurs when neither ICE nor EM2 provides power, allowing the vehicle to coast freely. During braking, **regenerative braking mode** activates, using EM2 as a generator to convert kinetic energy into electrical energy stored in the battery. In phases with low power demand, such as idling or coasting, the ICE is turned off and restarted by both electric motors when required, known as **start/stop mode**. When operational, EM1 precisely controls the ICE speed, enabling operation at the most efficient torque points (load point shifting), with excess mechanical energy stored in the battery. During acceleration, power from the battery can supplement the ICE through EM2, enabling higher overall output (**boost mode**). Generally, **EM1 functions as a generator, converting mechanical energy to electrical energy, which can directly power EM2**, be stored in the battery, or both. This advanced control allows power-split hybrids to effectively balance efficiency and performance across diverse driving scenarios.

Control Strategies Hybrid vehicles employ various control strategies to optimize fuel efficiency and performance. **Rule-based strategies** utilize predefined logic,

FIGURE 2 Illustration of a planetary gear set



often in the form of conditional statements based on vehicle speed, battery state of charge (SoC), and accelerator pedal position [8]. These strategies, common in production vehicles are robust, simple, and real-time capable, providing good performance when well-tuned. However, they lack foresight into upcoming driving conditions, resulting in consumption that can be 10–12 % higher than globally optimized strategies [9].

Fuzzy logic-based strategies allow smooth transitions between operating modes through intermediate truth values and can adapt to real-time conditions or driver behavior. Despite their adaptability, these strategies are less common due to increased complexity and reduced interpretability [8].

Optimization-based strategies include global and real-time optimization methods. Global optimization techniques, such as Dynamic Programming minimize cost functions based on known driving cycles but are not real-time capable. In contrast, real-time optimization methods like the Equivalent Consumption Minimization Strategy operate without drive cycle prediction, minimizing fuel and electrical energy consumption through an equivalence factor [9]. Additionally, reinforcement learning (RL) presents a promising advanced optimization approach, capable of adapting control strategies dynamically by learning optimal policies through continuous interaction with the driving environment. This adaptability makes RL particularly suitable for handling complex and varying driving conditions, further enhancing efficiency and performance [10].

Limitations of Conventional Simulation Methods

Conventional simulation tools, such as GT-SUITE by Gamma Technologies [11], are widely used for OD/1D modeling of engines and hybrid drivetrains. GT-SUITE is a multiphysics platform capable of simulating combustion, fluid flow, heat transfer, electrical components, and vehicle dynamics. The strength of such tools lies in their high modeling accuracy, provided they are properly calibrated and tailored to the specific application [12].

However, achieving high-fidelity results often requires solving complex thermodynamic and fluid-dynamic equations, which is computationally intensive. Detailed GT-SUITE models typically run much slower than real time, especially during transient driving cycles or when modeling large systems. To enable faster simulations, simplified models - so-called fast-running or mean-value models - are created by reducing system complexity [13]. While these models improve computation time, they sacrifice accuracy due to additional assumptions and the use of lookup tables. As a result, transient effects like inertia are no longer represented, which can be critical for specific use cases such as pre-turbine aftertreatment system modeling. As highlighted earlier, modern control and optimization approaches like reinforcement learning require thousands of simulations or rapid interaction episodes, which are infeasible with slow high-fidelity

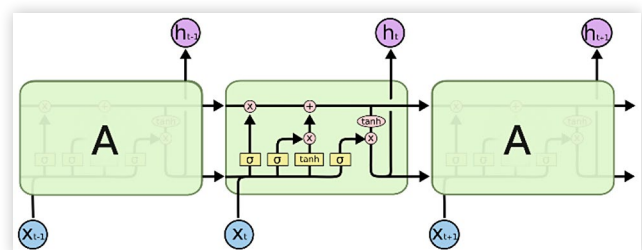
models. To address this, several studies have proposed replacing individual GT-SUITE models with neural networks to accelerate simulation [14]. In this work, a more comprehensive approach is taken: instead of replacing selected subsystems, the entire GT-based powertrain model is substituted. High-fidelity GT data is used to train neural networks externally, enabling the reconstruction of the full powertrain through an interconnected neural network architecture.

Data-Based Modeling in Automotive Systems

The growing complexity of powertrains and the increasing availability of high-resolution simulation and test data have made data-based modeling a valuable complement to traditional physics-based approaches. ANNs in particular, have proven to be powerful tools for capturing the nonlinear dynamics inherent in internal combustion engines and hybrid systems. A few studies – such as those focused on modeling gasoline combustion or predicting torque and tailpipe emissions – demonstrate the potential of ANNs in automotive applications [15] [16]. In hybrid vehicles, the prediction of battery State of Charge (SoC) is a common modeling task. While basic estimations can be performed using energy balance approaches, capturing the nonlinear behavior of the battery under dynamic and aging conditions can benefit from data-driven modeling [17]. NN-based approaches are used either as fast surrogate models for simulations or as components for real-time estimation and control.

Among various architectures, Long Short-Term Memory (LSTM) networks stand out for modeling temporal dependencies, which are particularly relevant for hybrid powertrains (Figure 3). As a subclass of recurrent neural networks, LSTMs incorporate gating mechanisms that enable the learning of long-term dependencies and time-lagged effects [18]. This is essential in automotive contexts where the system state - such as that of the engine, battery, or turbine - depends on sequences of past inputs and outputs. An LSTM cell contains three gates (input, forget, output) and two states (cell state and hidden state). The cell state acts as long-term memory, regulated by the gates. The input gate determines which new information is stored, the forget gate removes unnecessary information, and the output gate defines what information

FIGURE 3 Internal structure of a LSTM cell and the flow of information over time and cell from [19]



is passed on. The hidden state carries context from one timestep to the next. Together, these components allow the network to retain relevant information over long sequences while filtering out irrelevant data [19].

NNs can be trained on experimental or simulation data to learn system behavior directly, without requiring explicit physical or differential equations [20]. However, this black-box nature poses challenges for interpretability and validation, especially in cascaded structures. Unlike physics-based models, LSTMs do not inherently provide insights into the system and may behave unpredictably outside their training domain. Therefore, the quality and coverage of training data are critical. Networks can only learn what is present in the data, and errors such as outliers or measurement noise can degrade model generalization. Ensuring representative, clean, and comprehensive datasets is therefore essential for the development of robust neural network models.

Cascaded Neural Networks: Structure and Relevance

Cascaded neural networks refer to a structure of two or more neural networks that are arranged in a sequence, where the model output of one network serves as input for the next model. The overall prediction is broken down into stages or subtasks that are solved one after each other. In cascaded setups, the information flow is typically feed-forward through the stages [21]. In some setups, a feedback loop is required if one network is dependent on the output from a later network or if a timestep shift is required in time-dependent problems. Most applications of cascaded approaches can be found in robotics, computer vision, or anomaly detection [22] [23]. Only a few approaches exist for sequential data analysis with system modeling and simulation [24] [25]. A common example is time-series modeling, where a network's output at one timestep is reused as input for the following timestep. This self-feeding structure forms a feedback loop and requires special training methods such as teacher forcing, where predicted outputs are used as inputs during training. Each stage produces a final result for a sub-problem, consumed by the next network. In complex engineered systems, this enables a modular structure where different subsystems are modeled in sequence. For example, one network may predict an intermediate state, which another network then uses with additional inputs to predict the final output [21]. This decomposition aligns with how complex problems are typically approached and enables specialization in each subnetwork. It also allows replacing or improving a single model without affecting the entire structure – though this is most feasible if data generation followed a modular approach as well. This becomes challenging in transient systems where component interactions exist already during data generation.

In practical use, submodels are usually trained independently on dedicated datasets that may vary in structure, scale, or fidelity. Joint training across the full cascade

is rarely feasible, especially when datasets are not derived from a fully coupled simulation. This fragmentation can lead to mismatches during inference, where downstream networks receive statistically different inputs than they encountered during training. Traditional mitigation strategies like teacher forcing or end-to-end training are often not applicable in these scenarios.

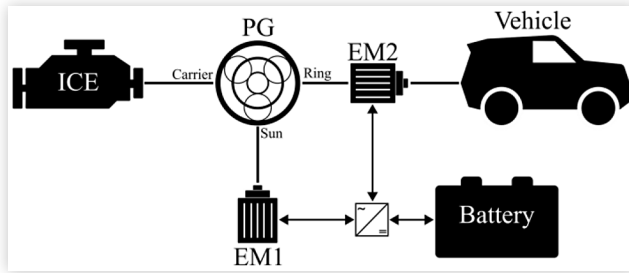
The main disadvantage of cascaded networks is error propagation. If one network produces a poor prediction, this flawed output feeds into subsequent models and may lead to compounding errors [24]. This is a well-known issue in sequential prediction tasks, such as multi-step forecasting, where error accumulation can lead to increasingly poor results over longer times. The mismatch between perfect training data and imperfect inference inputs exacerbates this problem, especially since neural networks typically lack physical constraints and can produce unrealistic outputs without any awareness.

Training methodology also plays a crucial role. While training all models together could yield better results by maintaining consistency between training and application flows, this requires unified, system-wide datasets – often making the cascaded approach unnecessary. Another challenge with cascaded networks for time-series data is the need for step-by-step predictions. Since each network relies on the previous network's output from the same timestep, it becomes necessary to predict sequentially rather than in batches, which can increase computational cost and reduce parallelism. While a single network prediction can operate faster than real time, the prediction performance of a cascaded structure may be reduced – potentially to the point where real-time applicability is no longer feasible, depending on the number of submodels and the available hardware.

Methodology

System Overview and Simulation Setup

Overview and Model Separation The simulation setup was developed in GT-SUITE. As described earlier, the power-split hybrid powertrain consists of an internal combustion engine (ICE), a planetary gearbox, two electric motors, a battery, and a vehicle model. A schematic illustration of the system architecture is shown in Figure 4. The core component of this study is the ICE, which is modeled in high detail. The 4-cylinder diesel engine includes combustion modeling, high- and low-pressure exhaust gas recirculation, turbine, compressor, and a comprehensive aftertreatment system – comprising SCR before the turbine, DOC, DPF, and SCR2 after the turbine. Due to the pre-turbine layout, precise temperature and thermal inertia modeling are essential, justifying the use of a detailed model despite its computational cost. This ICE model has been calibrated using test bench data and

FIGURE 4 Illustration of the complete powertrain setup

will not be further detailed in this work. Specifications of the engine can be found in [26].

The remaining powertrain components are modeled using map-based representations due to the absence of dedicated measurement data. Realistic parameterizations are based on in-house expertise, with component specifications provided in Table 1.

The complete powertrain model, especially in transient simulations, is computationally intensive due to interacting thermal and mechanical inertias. For instance, simulating a full WLTC of 1,800 seconds requires approximately 40 hours, corresponding to a real-time factor of around 90, depending on the specific case. Moreover, numerical stability is a challenge, as individual submodels have different boundary conditions and valid operating ranges. The in-house combustion model for example, is limited in simulating low-speed conditions but shows high accuracy in the valid range. Additionally, the aftertreatment models are sensitive to enthalpy spikes and abrupt mass flow variations, especially in the pre-turbine setup. These limitations can cause simulation failures and must be considered when defining the data generation setup.

To address computational limitations, the powertrain model is split into two submodels: a standalone ICE model

FIGURE 5 Numerical ICE model

(Figure 5) and a separate planetary gearbox–electric motor (PG-EM) model. This modularization provides several benefits: both submodels run significantly faster – particularly the PG-EM model – and can be parametrized over a wider range, enabling the generation of larger datasets. The ICE submodel can then be confined to valid operating regions, improving robustness and data quality.

TABLE 1 Component specifications of the planetary gear, electric system and vehicle

Component	Parameter	Value
Planetary Gear	Fixed Gear Ratio	-1.4
	Number of Ring Gear Teeth	87
	Number of Sun gear Teeth	62
	Number of Planetary Gears	4
EM1	max. Torque	115 Nm
	max. Speed	10,000 rpm
	Power	50 kW
EM2	max. Torque	421 Nm
	max. Speed	6,000 rpm
	Power	110 kW
Battery	Nominal Capacity	60 Ah
	Nominal Voltage	264 V
Vehicle	Empty Mass	1800 kg
	Payload	100 kg
	Drag Coefficient	0.36

Challenges of Numerical Model Simulations Unlike conventional simulations aimed at evaluating control strategies, the objective here is data generation for neural network training. As such, no rule-based controllers are applied. The overarching goal is to enable the neural network to learn and predict system behavior under the widest possible range of boundary conditions – as long as these are physically representable by the numerical model. Imposing predefined control strategies would limit this diversity, potentially restricting the model's generalization capabilities and its applicability in downstream optimization tasks. Instead, inputs to the ICE and PG-EM submodels are systematically varied according to predefined parameter sets and boundaries, as described later. The ICE model used for data generation is relatively simple and straightforward: fuel mass and engine speed serve as inputs, resulting in torque and emission outputs. In contrast, the PG-EM model presents additional complexity due to the planetary gear configuration.

In a planetary gear system consisting of sun, carrier, and ring shafts, only two degrees of rotational freedom exist because of the fixed kinematic constraint described by the Willis equation (2). The vehicle speed directly

determines the ring gear speed, effectively setting a boundary condition for the system. Unlike quasi-static approaches, which enforce torque balance at all times equations (3) and (4), dynamic simulations allow for torque imbalances that result in angular acceleration based on Euler's second law [27]:

$$\tau_i = J_i \cdot \dot{\omega}_i \quad (6)$$

With τ_i as net torque, J_i as shaft mass inertia and $\dot{\omega}_i$ describing angular acceleration.

This approach enables the assignment of torques to multiple shafts independently – e.g., prescribing ICE torque on the carrier (τ_c) and EM1 torque on the sun gear (τ_s) – without immediately violating the gear kinematics. However, any imbalance among τ_c , τ_s , and τ_r results in a net torque on the planetary system. This net torque accelerates the unconstrained shaft (typically the EM1 or ICE), leading to rapid and possibly unrealistic shaft speeds. In such cases, the torque relation from equation (5) is not enforced algebraically but only holds on average over time through the system dynamics. When the ring speed is fixed (i.e., at constant vehicle speed), torque imbalance will be resolved by accelerating the sun or carrier shaft, depending on the system's inertia distribution.

Without adequate control, this can lead to physically implausible outcomes such as ICE overspeed. To prevent this, an additional constraint is required to reduce the degrees of freedom in the Willis equation. In this work, this is achieved using a proportional-integral (PI) controller that regulates ICE speed by adjusting EM1 torque. As a result, the EM1 torque is no longer a direct input parameter. Instead, the system input is an ICE target speed, which is tracked through controlled adjustment of EM1 torque (Figure 6).

Data Generation and Preprocessing

To train transient-capable neural networks that can replace detailed numerical simulation environments, it is

essential to define a comprehensive and representative data generation strategy. The required search space is large and depends on several factors. First, the number of variation parameters plays a critical role. For the ICE model, this number has been minimized to four to reduce complexity. In contrast, the fast-running PG-EM model allows for a broader range of parameter variation, which is described in more detail below. Second, the system dynamics and their representation in the dataset are crucial. Each parameter exhibits unique transient behavior, which must be accurately modeled to reflect real system dynamics and support subsequent optimization tasks. Furthermore, the dataset must ideally cover the full parameter space, including all meaningful combinations. In a transient simulation, it is practically impossible to verify whether all interactions and dynamic transitions are fully covered. Therefore, test runs are typically used at the end of the training process to identify weak spots in the dataset. All used variation parameters and their range are shown in Table 2.

In this work, an automated framework was implemented to generate random time series for each variation parameter, defining the boundary conditions for all simulation runs. Two types of time series were used: normal ramp sequences and random jumps. Ramp sequences involve alternating periods of constant and ramping behavior. The ramp parameters (e.g., duration, gradient) are drawn from normal distributions, allowing for a wide range of dynamic behavior. This method ensures continuous and smooth transitions. Random jump sequences consist of constant periods followed by abrupt parameter changes to a new level, mimicking step-like transitions. These were applied to all control parameters except ambient conditions (which remained constant during each cycle) and the SoC, which only defined the cycle's initial condition.

Using this methodology, 93 simulation cycles were generated for the ICE model and 3,630 cycles for the PG-EM model, each with a duration of 1,800 seconds. The data generation process for the ICE model took approximately one month due to its computational intensity, while the PG-EM model simulations were completed

FIGURE 6 Numerical PG-EM model with PI controller

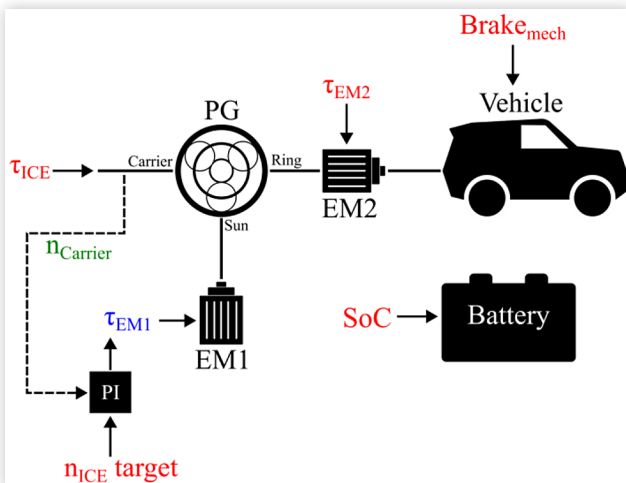


TABLE 2 Definition of relevant variation parameters for transient data generation and the specific parameter ranges

Variation Parameter	Operating Range
ICE model	
Ambient temperature (T_{amb}) [K]	[250; 310]
Ambient pressure (p_{amb}) [bar]	[0.8; 1.1]
Speed (n_{ICE}) [rpm]	[900; 4000]
Fuelmass (m_{fuel}) [mg]	[3; 70]
PG-EM model	
ICE Speed (n_{ICE}) [rpm]	[0; 5000]
ICE Torque (τ_{ICE}) [Nm]	[0; 500]
EM2 Torque (τ_{EM2}) [Nm]	[-422; 422]
Mechanical Brake ($Brake_{mech}$) [%]	[0; 100]
Start SoC (SoC) [%]	[0; 100]

within one week. Simulations were distributed across several nodes, each equipped with an Intel Core i9-11900K 8-Core 3.5 GHz processor. At any given time, it was possible to run between 6 and 9 simulation cycles in parallel, depending on overall system availability.

This raw simulation data was then preprocessed to prepare it for neural network training. All timeseries were **resampled to 2 Hz**. This frequency was deemed sufficient to capture the essential dynamics of key system outputs such as **speed, torque, and SoC, which typically evolve at relatively moderate rates**. While 2 Hz proved adequate for these parameters, it is acknowledged that **a more detailed frequency analysis could potentially improve the prediction quality of faster-reacting variables, such as emissions**. Nevertheless, the chosen sampling rate significantly reduced computational effort during training and enabled faster-than-real-time inference in the application phase. Outliers were removed to improve data quality and avoid misleading the learning process. The data was then scaled using a Min-Max scaler with a range of $[-1; 1]$ to minimize the influence of differing units and numerical magnitudes among the input features. Without scaling, features with large numeric ranges may dominate the learning process, particularly in gradient-based optimization. Min-Max normalization ensures balanced feature contributions and prevents numerical instability. The $[-1; 1]$ range was specifically chosen to match the domain of activation functions like tanh, which are common in recurrent neural networks such as LSTM used in this work. Finally, the data was randomly split into training, validation, and test sets. For the ICE model, the dataset was split into 70 % training, 20 % validation, and 10 % test. In contrast, the PG-EM model used a split of 50 % training, 30 % validation, and 20 % test. These differing ratios reflect the varying amounts of data available for each model. In the case of the ICE model, a larger portion of the dataset was allocated to training to ensure that the neural network was exposed to all relevant system behaviors during learning. This was particularly important given the more limited dataset size and the need to cover the full operational range of the ICE.

Neural Network Architecture and Training Procedure

Designing an appropriate neural network architecture is a complex and application-specific task. As discussed earlier, LSTM cells are well-suited for time-series prediction, especially in highly nonlinear systems like powertrain applications. For this reason, LSTM networks were selected in this work. Choosing the exact architecture and tuning the hyperparameters requires a combination of expert knowledge, experience, and iterative experimentation. Increasing the number of hidden layers enhances the model's capacity to learn complex patterns, but it also raises the risk of overfitting – particularly when training data is limited. Overfitting occurs when the model memorizes the training data instead of learning generalizable rules [28]. There is no universal equation for the optimal

TABLE 3 Neural network architectures and hyperparameters used for training and final test loss

Parameter	ICE	PG-EM v1	PG-EM v1-shift	PG-EM v2
Layers	3	3	3	3
Neurons	128, 128, 128	256, 256, 256	256, 256, 64	256, 256, 256
Sequence length	300	3600	300	3600
Batchsize	1	1	1	1
Optimizer	AdamW	AdamW	AdamW	AdamW
Initial Learning rate	1e-3	1e-3	1e-3	1e-3
Scaler	Min-max-scaler	Min-max-scaler	Min-max-scaler	Min-max-scaler
Loss function	MAE	MAE	MAE	MAE
Test Loss (MAE)	0.016	0.035	0.105	0.018

ratio between training data and model complexity, as it depends on various factors such as task complexity, data noise, model architecture, and regularization techniques [29]. The final network configurations used in this work are described in Table 3.

A critical aspect of working with LSTM networks is the initialization of hidden and cell states at the beginning of each prediction cycle. By default, these states are initialized to zero, which can cause a short warm-up phase with reduced prediction accuracy. To address this issue, the method proposed by Hagenbucher et al. [30] was used. This technique involves training auxiliary layers to generate suitable initial states based on the first output values of each sequence. In this work, the method was successfully applied, improving prediction quality from the beginning of each simulation cycle.

Cascaded Neural Network Structure

As described at the beginning of this work, the adoption of a cascaded neural network structure is motivated by multiple factors. In the present case, the motivation stems from the data generation approach. By splitting the numerical models into two components – one for the ICE and one for the PG-EM system – it was possible to significantly increase the number of data points within the same computational time and to expand the variation range of the PG-EM model inputs, as these are controlled directly.

Based on this separated data generation, two primary models were developed and the subnetworks were trained individually: the ICE neural network (NN_{ICE}), the PG-EM v1 neural network ($NN_{PG-EM v1}$), the PG-EM v1-shift neural network ($NN_{PG-EM v1-shift}$) and the PG-EM v2 neural network ($NN_{PG-EM v2}$). The following section focuses on the

design of the cascaded structure itself – specifically, how these networks are interconnected and what assumptions and constraints must be considered to ensure consistent and physically meaningful predictions across the full hybrid system.

Cascaded Network Model 1 The development of the first cascaded neural network model follows a logical and physically consistent approach. The overall structure is divided at the mechanical interface between the ICE and the planetary gearbox, specifically at the common shaft connecting the ICE to the carrier gear. This mechanical interface naturally defines torque and rotational speed as the coupling variables between the two subnetworks. The NN_{ICE} predicts the torque output of the internal combustion engine based on m_{fuel} , T_{amb} and p_{amb} and n_{ICE} . The $NN_{PG-EM v1}$ and $NN_{PG-EM v1-shift}$ then takes the predicted τ_{ICE} as a cascaded input, along with τ_{EM1} , τ_{EM2} , and the Brake_{mech} input, to predict downstream behavior such as vehicle speed (v_{car}), battery state of charge (SoC), and, crucially, the carrier speed at the next timestep ($n_{Carrier}$). A representation of the described model is shown in Figure 7.

A key feature of this architecture is the timestep shift for the predicted $n_{Carrier}$. At timestep $t=0$, the NN_{ICE} computes the τ_{ICE} based on the current conditions. This torque is then used as an input for the $NN_{PG-EM v1-shift}$, which predicts the carrier speed at $t+1$. This shift reflects the dynamic response of the planetary gear system, where torque inputs at one timestep result in accelerations and new shaft speeds in the next as described in section ‘Challenges of numerical model simulations’.

During data generation, a PI controller was used to maintain n_{ICE} by adjusting τ_{EM1} . While τ_{EM1} inherently contains the influence of the original controller used during data generation, it was directly used as an input feature for $NN_{PG-EM v1}$ and $NN_{PG-EM v1-shift}$ instead of the controller target. This ensures that the neural network does not learn the controller behavior itself, preserving the flexibility to apply alternative control strategies later on.

This modular and temporally cascaded setup allows for a physically interpretable and simulation-compatible model. It preserves transient dynamics and enables

flexible integration in downstream optimization tasks, such as reinforcement learning-based control.

Cascaded Network Model 2 While the first cascaded model was physically consistent, two major limitations emerged during application and testing. First, the prediction of the $n_{Carrier}$ proved to be highly unreliable. Second, when deployed in a closed-loop cascade, the model exhibited pronounced oscillations, with output predictions fluctuating strongly around the real values. These effects are analyzed in more detail in the ‘Results and Discussion’ section.

To overcome these issues, a modified network structure was developed. The central challenge stemmed from the high sensitivity of the carrier speed to small torque changes. Predicting the carrier speed accurately based solely on torque inputs – without considering the current or target n_{ICE} – proved too difficult for the neural network. To address this, a new assumption was introduced: it is assumed that the PI controller used during data generation reliably enforces the n_{ICE} target. In each simulation run, the τ_{EM1} was dynamically adjusted so that the n_{ICE} closely tracked the predefined target speed. In fact, over 75 % of all simulation cycles showed a controller error below 25 rpm, confirming the validity of this assumption.

Based on this observation, the n_{ICE} -target was used directly as a model input. Therefore, the training dataset was filtered to exclude cycles with larger control errors. This ensured that the model only learned under conditions where the n_{ICE} and its target were well aligned, resulting in the model shown in Figure 8.

As a result, the internal feedback loop from Model 1 could be eliminated. Instead, the n_{ICE} becomes a shared, externally defined input to both the NN_{ICE} and NN_{PG-EM} . Functionally, the control task of EM1 is implicitly absorbed into the NN_{PG-EM} model: the n_{ICE} target is treated as a known boundary condition, and τ_{EM1} no longer needs to be predicted or specified.

This revised architecture improves model stability and prediction robustness. By eliminating the internal dynamic feedback and replacing it with a stable, predefined reference input, the model becomes less sensitive, avoids oscillatory behavior, and generalizes more reliably across unseen conditions. Naturally, this assumption introduces a limitation: the model no longer explicitly learns the τ_{EM1} response or controller dynamics, which may reduce flexibility in scenarios where the controller behaves

FIGURE 7 Representation of Cascaded Network Model 1

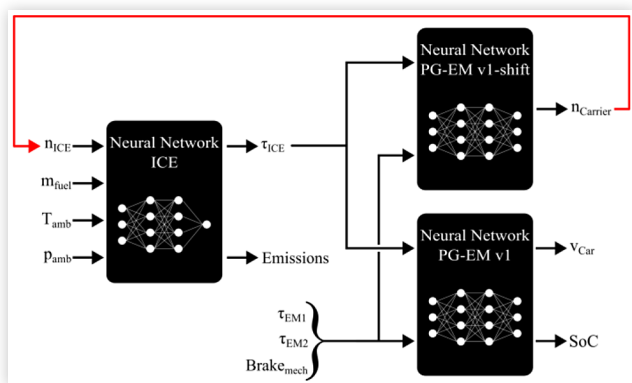
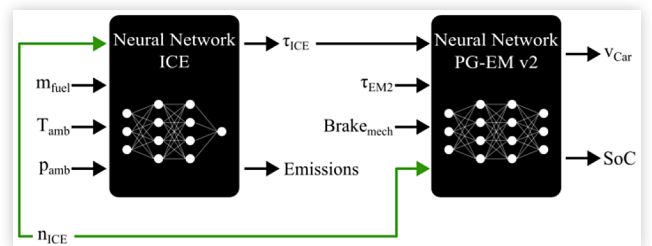


FIGURE 8 Representation of Cascaded Network Model 2



suboptimal or operates near its limits. However, in the context of this work, the assumption was considered the only practical and effective way to resolve the observed prediction instability and feedback-induced oscillations.

Results and Discussion

Overview of Evaluation Approach

Evaluating a cascaded neural network model requires careful consideration of both its individual components and the integrated system. The performance of each subnetwork must be analyzed separately, while also assessing how errors propagate through the cascade. This dual perspective is essential for understanding not only local prediction quality but also the robustness of the overall system.

To quantify prediction accuracy, two evaluation metrics are used: the Mean Absolute Error (MAE) (equation (7)) and the Mean Absolute Percentage Error (MAPE) (equation (8)). The MAE calculates the average absolute difference between the predicted and actual values:

$$MAE = \frac{1}{n} \sum_{i=1}^n |x_{pred} - x_{real}| \quad (7)$$

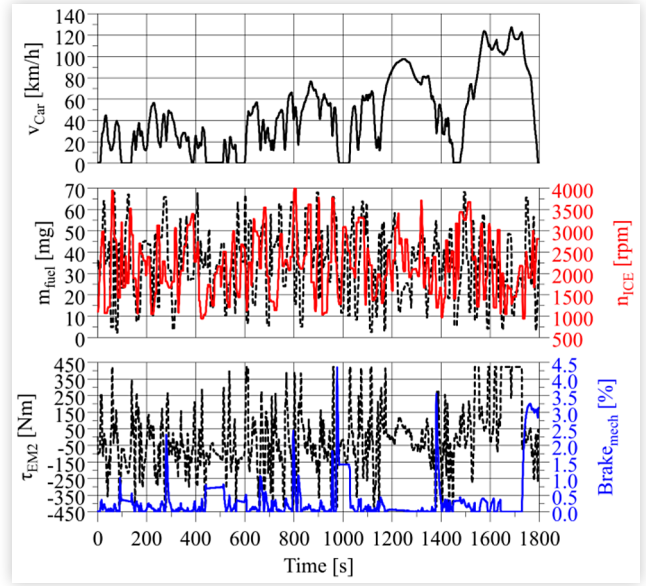
The MAPE expresses this difference as a percentage of the real value:

$$MAPE = 100 \frac{1}{n} \sum_{i=1}^n \left| \frac{x_{pred} - x_{real}}{x_{real}} \right| \quad (8)$$

Here, x_{pred} refers to the network prediction and x_{real} to the corresponding value from the numerical GT-SUITE simulation. In the following sections, “real values” are referred to simulation data. Since MAPE becomes enormously large when the real values approach zero, both metrics are used in parallel to ensure robust and interpretable model evaluation. All output variables of the networks are evaluated independently, with performance assessed as an average over a complete transient test cycle.

To ensure a reliable evaluation, it is critical that the test data has not been used during training or validation. In this work, the WLTC is selected as the evaluation benchmark. This standardized cycle is widely used in automotive development and offers a representative transient driving scenario. A simplified but realistic control strategy was used to generate the test scenario. ICE torque and speed signals were randomly chosen from a held-out test case and applied to the PG-EM model. EM1 was regulated to track the ICE speed target, while EM2 and the mechanical brake were controlled to follow the WLTC vehicle speed. The initial SoC was set to 20 % and ambient conditions were a temperature of 286 K and a pressure of 0.96 bar. The cycle with the relevant input parameters is shown in Figure 9.

FIGURE 9 WLTC testcase input parameters



The next section presents the results obtained using this evaluation framework and analyzes the model's accuracy, stability, and generalization performance. The plots presented in the following sections often focus on zoomed-in segments of the WLTC cycle for improved clarity and detailed analysis. Full-length cycle plots are provided in the appendix.

Model 1: Neural Network Model Evaluation

For the modeling approach presented in this work, several standalone neural networks are required. Cascaded Model 1 utilizes three networks, while Cascaded Model 2 is composed of only two. As previously outlined, the PG-EM network is responsible for multiple prediction tasks. It must predict v_{Car} and SoC at the current timestep t , as well as the $n_{Carrier}$ at the next timestep $t+1$. To handle this, two distinct neural networks are employed: one for the predictions at timestep t , and a second for the prediction of $n_{Carrier}$. The third network involved is the NN_{ICE} , which models the internal combustion engine behavior. This one is the same network for both Model approaches.

Single Neural Network Performance

NN_{ICE} Single Prediction Analysis. Starting with NN_{ICE} which is the same for both models and therefore in the single evaluation only discussed here. The NN_{ICE} model demonstrates high accuracy in predicting all key outputs over the full WLTC cycle (Figure 10). In addition to the τ_{ICE} output required for the cascaded setup, the prediction of relevant tailpipe emissions is included as part of the network's output. The underlying numerical model applies a constant, map-based aftertreatment control strategy. A detailed analysis of this strategy is not provided here, as it lies outside the scope of this work. Predicted values (dashed

FIGURE 10 Single predictions of NN_{ICE} compared to real values

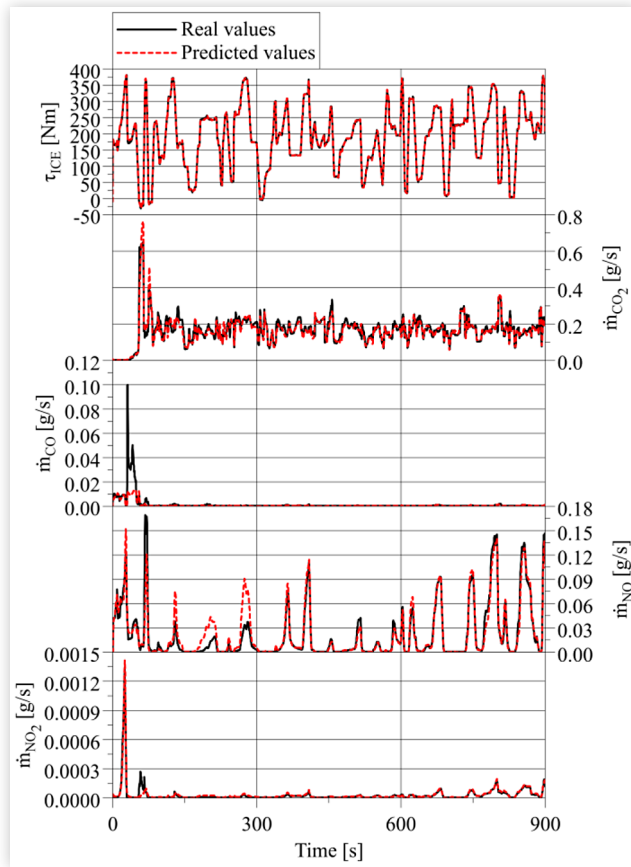


TABLE 4 Evaluation metrics of single NN_{ICE} predictions

Output Parameter	MAE	MAPE
τ_{ICE}	1.60 Nm	3.01 %
$\dot{m}_{CO_2}$	0.015 g/s	15,891.16 %
$\dot{m}_{CO}$	0.00045 g/s	367,412.78 %
$\dot{m}_{NO}$	0.0042 g/s	136.72 %
$\dot{m}_{NO_2}$	8.35e-6 g/s	159.72 %

red) closely follow the reference simulation values (solid black) across all five output channels, indicating that the network has successfully learned the input-output relationships present in the training data. Among the predicted quantities, torque prediction is especially accurate due to its strong correlation with the m_{fuel} input. As such, this parameter is comparatively easy for the network to learn.

The network captures both steady-state and transient dynamics with minimal phase shift, successfully reproducing peaks and drops throughout the cycle. Some minor under- and overshooting occurs during rapid transitions, particularly around 300 s and 800 s, but these errors remain bounded and do not affect the overall quality of the prediction.

From an engineering perspective, the prediction of tailpipe emissions is particularly noteworthy. Emissions arise from a chain of complex processes – combustion,

gas exchange, and catalytic conversion – all of which are influenced by thermodynamic states, control strategies, and transient operation. These dependencies introduce strong nonlinearity, and relevant intermediate states are only implicitly present in the input data. Nevertheless, the network delivers highly accurate emission predictions across the cycle. Minor deviations are visible throughout, but the most significant discrepancy occurs in the early part of the WLTC cycle for CO emissions. Here, while the predicted trend remains correct, absolute values deviate from the reference. This is likely due to complex effects taking place during cold-start or low-load effects, which result in incomplete combustion and catalyst inactivity – phenomena the amount of training data might not be sufficient. The metrics, especially the MAE, in Table 4 confirm the good accuracy with small errors overall.

$NN_{PG-EM v1}$ Single Prediction Analysis. This model is used to predict v_{Car} and SoC based mainly on the three torque inputs: τ_{ICE} , τ_{EM1} , τ_{EM2} . No shaft speed is provided as an input. The model results are shown in Figure 11.

The predictions show only moderate accuracy. The vehicle speed is both under- and overestimated at various points in the cycle (around 400s and 1,700 seconds). This can likely be attributed to the kinematics of the planetary gear, which must be learned implicitly by the network. Since no explicit speed information is included, the model must infer shaft accelerations and their interdependencies from torque inputs alone – following the physical relations described in equations (2), (5) and (6). This is particularly challenging due to the sensitivity of shaft acceleration to small torque changes, especially during frequent load shifts. More on that in the next section.

Among the three shafts, the ring gear (which is mechanically linked to the wheels via the differential) exhibits the slowest dynamic response. This likely explains why the vehicle speed prediction is somewhat reasonable in trend but still shows error. A similar effect is observed in the SoC prediction, which displays a rising offset over time. However, overall trends are reasonable and results are still useful for the shown time period. The accuracy in metrics show reasonable results, comparing Table 5.

FIGURE 11 Single predictions of $NN_{PG-EM v1}$ compared to real values

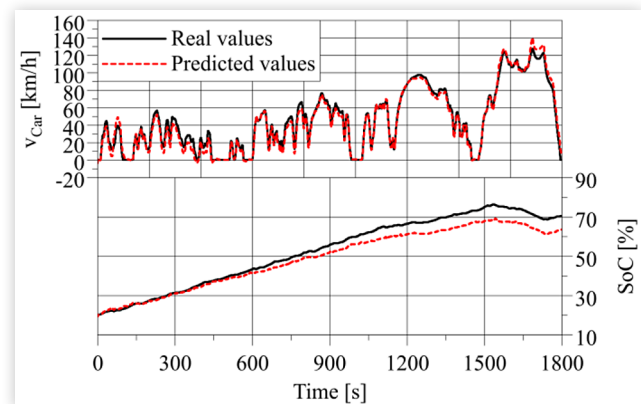


TABLE 5 Evaluation metrics of single $NN_{PG-EM\ v1}$ predictions

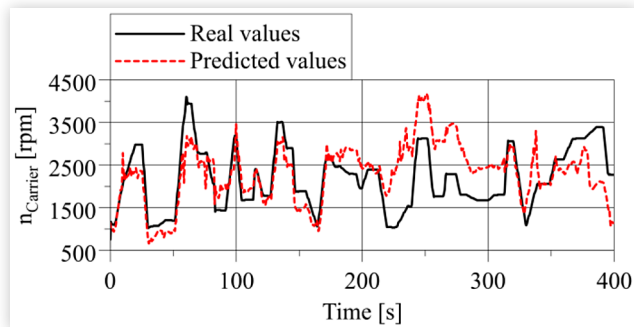
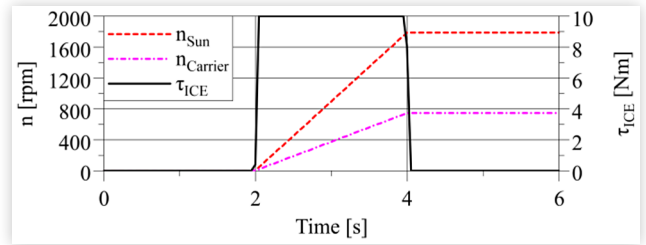
Output Parameter	MAE	MAPE
V_{Car}	3.58 km/h	15,209.31 %
SoC	3.67 %	6.01 %

$NN_{PG-EM\ v1}$ -shift Single Prediction Analysis. The neural network in Model 1 is tasked with predicting the ICE/Carrier shaft speed at the next timestep to enable the transient behavior of the cascaded structure. The standalone prediction results of this model are relatively poor, with an MAE of 507.43 rpm (Table 6). While the network reacts to torque changes with some accuracy in terms of timing and direction, it consistently fails to reproduce the correct absolute values of the resulting speed. A cutout is shown in Figure 12. The network is forced to implicitly learn the planetary gear kinematic constraints (equation (2)), which involve two degrees of freedom – typically the rotational speeds of the sun and carrier shafts – while the ring gear speed is fixed by the vehicle velocity through the differential. These degrees of freedom define how applied torques are dynamically distributed, and how resulting speeds evolve across the three shafts. Since the carrier and sun can rotate independently but are mechanically coupled through the gaset, even small mismatches in torque can lead to highly sensitive and nonlinear acceleration behavior. This is especially difficult to learn accurately when no explicit shaft speed is available as an input.

This sensitivity is clearly illustrated in the simulation shown in Figure 13. In this test case, both τ_{EM1} and τ_{EM2} are set to zero, and a constant torque of just 10 Nm is applied by the ICE for two seconds. Even under this simple condition, the carrier speed accelerates from 0 rpm to 742 rpm – demonstrating the system's extreme sensitivity to even small torque inputs. Such sharp and nonlinear responses present a major challenge for a data-based model when only torque values are provided as input features.

TABLE 6 Evaluation metrics of single $NN_{PG-EM\ v1}$ -shift predictions

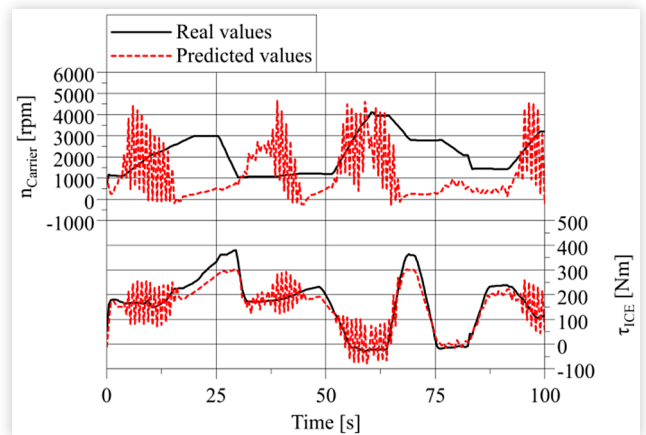
Output Parameter	MAE	MAPE
$n_{Carrier}$	507.43 rpm	25.63 %

FIGURE 12 Single predictions of $NN_{PG-EM\ v1}$ -shift compared to real values**FIGURE 13** Sensitivity analysis of $n_{Carrier}$ and n_{Sun} on τ_{ICE} 

This limitation is further supported by the training behavior of the network. Despite extensive tuning and multiple attempts, the model showed poor convergence during training (Table 3). The loss plateaued early and remained relatively high, suggesting that the network was unable to learn a consistent mapping from torque inputs to the correct carrier speed output. This confirms that the network is not just imprecise during inference, but fundamentally unable to resolve the dynamic relationship based on the given inputs.

Unlike numerical models, which explicitly solve the second Euler equation (6) using known inertia and torque values, the neural network lacks both this physical formulation and the complete input set required to approximate it. As a result, the model cannot reproduce the physical update of shaft speed based on torque balance and inertial dynamics. Without access to the current carrier speed or an internal integration mechanism, the network is structurally underdetermined and unable to replicate the correct dynamic behavior – particularly in transient torque conditions.

Cascaded Model 1 Performance As the individual predictions already suggest, the overall performance of the cascaded model is poor. The results in Figure 14 show not only inaccurate carrier speed prediction but also oscillatory behavior in τ_{ICE} . Only the first 100 seconds of the test cycle are displayed due to the high frequent oscillation. Despite the oscillations affecting the torque signal, the mean deviation remains small – possibly due to the

FIGURE 14 Model 1 cascaded predictions compared to real values between 0 s and 100 s

high dependence of diesel engine torque on fuel mass rather than shaft speed. Still, without external damping, the signals lack practical usability. The oscillating behavior propagates through the network cascade and affects all connected subsystems. In the described configuration, the predicted engine torque is fed into the $NN_{PG-EM\ v1-shift}$, which in turn predicts the carrier speed. This predicted speed then influences subsequent predictions, forming an implicit feedback loop. In such a structure, even small prediction errors can be amplified: if the model slightly overshoots in one timestep, it may overcorrect in the next, leading to sustained oscillations. Unlike physical simulations, neural networks do not inherently include damping, inertia, or stability constraints. Conventional numerical models account for such effects through physical laws, whereas a neural network must learn them implicitly from data. If the network is trained on "clean" data (i.e., ideal simulation outputs), but during application receives imperfect inputs (e.g., from preceding networks), the learned physical relationships may no longer apply – leading to unstable and unpredictable behavior. In this case, the feedback loop appears to result in positive error amplification. A small deviation in the carrier speed prediction alters the downstream inputs, which in turn lead to even larger errors in the next timestep. Without an internal mechanism to resist these deviations the network begins to under- and overshoot the true value and may even diverge. Interestingly, periods of high-frequency oscillations are often followed by relatively stable and accurate predictions. Despite the instability in the carrier speed output, the oscillatory behavior does not appear to significantly affect the performance of the $NN_{PG-EM\ v1}$ overall. As shown in Figure 15, the predicted v_{Car} remains relatively accurate and free of oscillations. This observation highlights an important aspect of the system: not all outputs are equally sensitive to all inputs. While the predicted τ_{ICE} is more dependent on $n_{Carrier}$, the vehicle speed is evidently less affected by these fluctuations. From a powertrain engineering perspective, this is plausible – since v_{Car} is primarily governed by the ring gear, which is mechanically linked to the wheels and exhibits much higher inertial resistance and damping. Even though the carrier speed oscillations are not directly reflected in v_{Car} , the compounded inaccuracies in the cascaded network indirectly reduce the fidelity of the vehicle speed prediction. The values of MAE and MAPE are shown in Table 7.

FIGURE 15 Model 1 cascaded v_{Car} prediction compared to real values

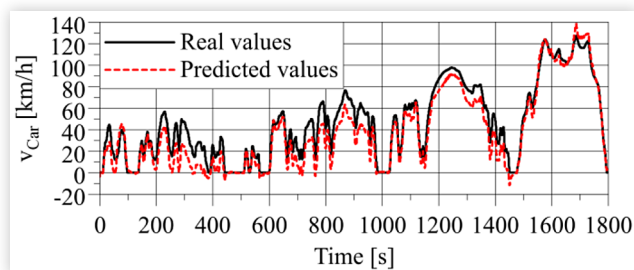


TABLE 7 Evaluation metrics of cascaded Model 1 predictions

Output Parameter	MAE	MAPE
τ_{ICE}	45.69 Nm	92.72 %
$\dot{m}_{CO2}$	0.051 g/s	133,347.91 %
$\dot{m}_{CO}$	0.0016 g/s	10,636,515.64 %
$\dot{m}_{NO}$	0.022 g/s	1,949.98 %
$\dot{m}_{NO2}$	4.10e-5 g/s	2,288.55 %
v_{Car}	9.13 km/h	30,545.48 %
SoC	6.35 %	10.71 %
$n_{Carrier}$	1544.18 rpm	72.79 %

Addressing this issue remains an open challenge. One potential solution is to introduce a mixture of real and predicted data during training, which helps the network adapt to its own prediction errors. However, this would require joint training and a joint database, which would call the cascaded approach into question. Another direction is to implement active damping externally during application.

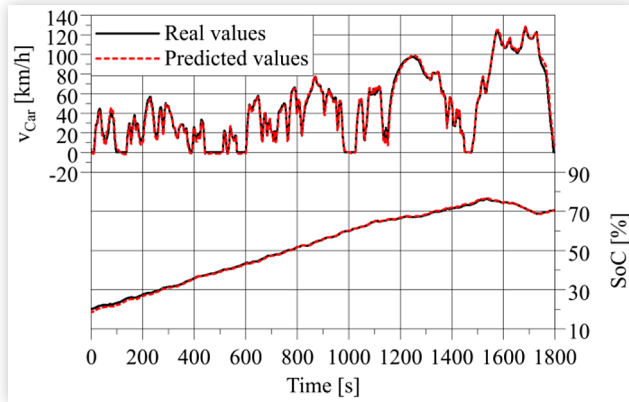
However, since the $NN_{PG-EM\ v1-shift}$ failed to converge during training, it is not surprising that the corresponding cascaded structure also performs poorly. To address this limitation, an alternative modeling approach was developed. As previously discussed, a key assumption was introduced: it was assumed that the PI controller used during data generation always reaches the n_{ICE} target. Based on this, the neural networks were provided with the n_{ICE} as an additional input, rather than relying solely on torque values.

This modification has two important effects. First, it simplifies the learning task by providing the necessary kinematic reference that the previous model lacks. Second, it eliminates the need for a feedback loop, since the ICE speed is now treated as a given input rather than a predicted output. The resulting architecture is described in the following section as Model 2.

Model 2: Neural Network Model Evaluation

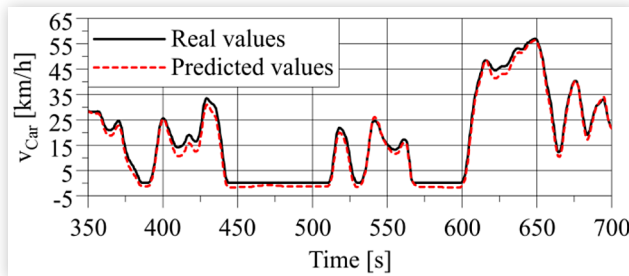
As described above Model 2 only consists of two networks. The NN_{ICE} is the same as above and therefore not further discussed in the single prediction evaluation here. The $NN_{PG-EM\ v2}$ has changed slightly by the input parameters.

Single Neural Network Performance $NN_{PG-EM\ v2}$ Single Prediction Analysis The $NN_{PG-EM\ v2}$ exhibits strong prediction accuracy across the WLTC cycle. Both the vehicle speed and the battery's state of charge are predicted with high fidelity (Figure 16). The predicted curves (dashed red) align very closely with the reference values (solid black) across all time steps.

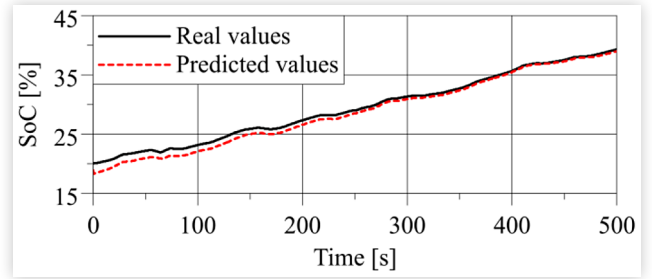
FIGURE 16 Model 2 $NN_{PG-EM\ v2}$ predictions compared to real values

The WLTC velocity profile is highly dynamic, consisting of low-, medium-, high-, and extra-high-speed phases. It includes frequent acceleration and deceleration segments, all of which are captured with high accuracy by the $NN_{PG-EM\ v2}$. The model reliably reproduces sharp transitions, cruise segments, and stop phases, with minimal time lag or overshooting. This demonstrates that the network has successfully learned the dynamic system behavior under varying torque and braking inputs, despite the absence of explicit vehicle dynamics equations. The biggest deviations occur when the vehicle stops. The prediction in that parts are slightly too low or high as shown in the zoom-in in Figure 17. This also reflects the large MAPE in Table 8.

The SoC prediction is particularly notable for its smoothness and precision, confirmed by a MAPE of 1%. Over the full 1,800-second cycle, the predicted SoC trajectory closely matches the simulated real values, with only negligible errors during the initialization phase. This can be seen in Figure 18. The overall charge-depletion and charge-recovery trends are well reproduced, and charging/discharging dynamics are accurately captured. This

FIGURE 17 $NN_{PG-EM\ v2}$ prediction compared to real values zoomed in**TABLE 8** Evaluation metrics of single $NN_{PG-EM\ v2}$ predictions of model 2

Output Parameter	MAE	MAPE
V_{Car}	1.99 km/h	33,370.54 %
SoC	0.35 %	1.00 %

FIGURE 18 $NN_{PG-EM\ v2}$ prediction compared to real values zoomed in

accuracy is particularly important for control applications such as energy management, where even small cumulative errors in SoC prediction could degrade system performance over time.

No significant phase shift, drift, or systematic bias (over- or underestimation) was observed, suggesting that the $NN_{PG-EM\ v2}$ model generalizes well across a broad range of hybrid drivetrain operating points. This consistency indicates that the training data was sufficiently diverse to capture the essential system dynamics and that the chosen network architecture is well suited to this regression task.

Cascaded Model 2 Performance The cascaded structure of Model 2 demonstrates performance comparable to that of the individual networks. Since no visual difference can be observed, a plot is omitted at this point. As outlined earlier, in contrast to Model 1, this approach integrates the PI controller into the $NN_{PG-EM\ v2}$ implicitly by using the τ_{ICE} as an input. This simplification enables τ_{ICE} to serve as the only transferred coupling parameter between the subnetworks.

Evaluating the outcome of the cascaded predictions shows that overall prediction accuracy remains unchanged compared to isolated single-network performance. A detailed inspection of Table 9 confirms that the NN_{ICE} output errors remain constant, while the vehicle speed error increased by only 0.03 km/h – an effect considered negligible. Interestingly, the SoC prediction even improved slightly, with a MAPE reduction of 0.07 %. While this improvement may be coincidental, it indicates that the cascaded setup does not compromise predictive accuracy.

TABLE 9 Evaluation metrics of cascaded Model 2 predictions

Output Parameter	MAE	MAPE
τ_{ICE}	1.49 Nm	2.81 %
$\dot{m}_{CO2}$	0.015 g/s	13,874.52 %
$\dot{m}_{CO}$	0.00045 g/s	364,325.16 %
$\dot{m}_{NO}$	0.0042 g/s	137.04 %
$\dot{m}_{NO2}$	8.30e-6 g/s	159.09 %
V_{Car}	2.02 km/h	37,702.02 %
SoC	0.33 %	0.93 %

A direct comparison of the errors between Model 1 and Model 2 can be found in Table 10 in the Appendix section.

The consistency of the τ_{ICE} prediction plays a crucial role here. As the only link between NN_{ICE} and $NN_{PG-EM\ v2}$, any error accumulation would propagate through this variable. However, since prediction accuracy remains stable, it confirms the robustness of the cascaded approach and suggests that no significant error amplification occurs during the inter-network transfer in the present case.

Important Findings and Summary

While these findings are encouraging, it is important to note that all statements made above are specific to the current system and configuration. As the instability observed in Model 1 demonstrates, cascaded neural network architectures do not universally work for all applications or under all conditions. Their suitability must be evaluated for each specific case.

Numerous influencing factors are at play – most importantly, the characteristics of the underlying system. In this work, the power-split hybrid drivetrain with a planetary gear introduces considerable complexity, especially because the cascading in time-series applications requires the successful transfer of a parameter across timesteps. In this context, the dynamic shift could not be reliably predicted by the network in Model 1 due to the high sensitivity and coupled dynamics of the system. However, this issue may not exist in simpler systems or different architectures.

On the neural network side, it is also important to consider that not all inputs have equal influence on the prediction targets, as observed in Model 1. In Model 2, only a single variable (τ_{ICE}) was passed between networks, while other input parameters of the PG model were externally provided. These additional inputs can act as stabilizers, limiting the propagation of error. Moreover, if the ICE torque has a relatively weak influence on downstream outputs such as vehicle speed, even moderate errors may have minimal overall impact. These dependencies must be carefully considered in the design and application of cascaded network architectures.

A particularly noteworthy aspect is the prediction time required to run the data-driven models. Since each neural network in the cascaded structure was trained independently, each has its own input and output scalars. During inference, the process is repeated for every timestep: the input data is first scaled, the prediction is performed, and the output is then unscaled before being passed to the next network in the cascade. For a single LSTM network, the prediction time for one timestep is approximately 0.001 seconds on an Intel i7-13700K CPU. Running the full WLTC cycle (1800 seconds, sampled at 2 Hz) takes about 14.04 seconds for the complete cascaded structure. In comparison, the equivalent numerical simulation requires approximately 38 hours – resulting in a massive reduction in computation time. It is important to note that this comparison only considers the runtime

of the trained models, not the time required for initial data generation. Additionally, the prediction time of the data-driven model scales with the number of subnetworks: more networks in the cascade naturally increase total runtime. The real-time factor of the numerical simulation is approximately 76, whereas cascaded Model 2 achieves a real-time factor of 0.0078, meaning it runs about 128 times faster than real-time.

Conclusions

This work presented a data-based modeling approach for a power-split hybrid powertrain using cascaded neural networks. Two different network structures were developed and evaluated to replace a detailed numerical simulation environment with a modular, machine learning-based alternative. The models were trained on transient data generated via GT-SUITE and validated against the WLTC driving cycle. The first approach (Model 1) followed a physically intuitive structure by predicting the carrier speed dynamically as a function of torque inputs. However, due to the high sensitivity of the planetary gear system and the lack of explicit speed references, the PG-EM v1-shift network failed to converge during training. Consequently, the full cascaded setup suffered from oscillatory behavior and poor accuracy, especially under feedback conditions. These findings demonstrate the structural limitations of networks attempting to learn underdetermined physical relationships without access to essential kinematic states. To overcome this, Model 2 introduced a practical assumption: that the internal PI controller reliably enforces the ICE speed target. This allowed the ICE speed to be used directly as a network input, breaking the feedback loop and significantly improving prediction stability and accuracy. The cascaded Model 2 showed consistent performance across all outputs, with MAE and MAPE values demonstrating good agreement with simulation results. Furthermore, inference time was reduced from 38 hours in the numerical model to 14.04 seconds in the data-driven model, enabling faster-than-real-time execution.

Despite the promising results, several limitations remain. The neural networks lack inherent physical constraints, which can lead to error amplification in open-loop feedback configurations. Additionally, training was performed on clean simulation data, which does not necessarily reflect noisy or suboptimal signals. The need for assumptions (e.g., ideal controller performance) also limits generalizability.

In addition to architectural improvements, future work should also consider refining the training pipeline. More comprehensive hyperparameter optimization could further improve accuracy and stability. Likewise, the sampling frequency of the training data may have a strong impact on the model's ability to capture fast transient dynamics. While 2 Hz was used in this study, increasing the resolution could allow the networks to better resolve rapid changes in emissions, torque and

speed. Systematic investigation of preprocessing methods, feature scaling strategies, and frequency-dependent effects could reveal additional performance gains.

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Definitions/Abbreviations

ANN - Artificial Neural Network

ICE - Internal Combustion Engine

DOC - Diesel Oxidation Catalyst

DPF - Diesel Particulate Filter

EM1/EM2 - Electric Motor 1 / Electric Motor 2

GT-SUITE - Gamma Technologies Simulation Software

LSTM - Long Short-Term Memory

MAE - Mean Absolute Error

MAPE - Mean Absolute Percentage Error

NN - Neural Network

PI - Proportional-Integral (Controller)

PG-EM - Planetary Gear - Electric Motor (model)

RL - Reinforcement Learning

SCR - Selective Catalytic Reduction

SoC - State of Charge

WLTC - Worldwide Harmonized Light Vehicles Test Cycle

i_{fix} - Fixed Gear Ratio

$\dot{m}$ - Mass Flow Rate

m_{fuel} - Injected Fuel Mass per Cylinder

$n_{carrier}/n_c$ - Carrier Shaft Speed

n_{ICE} - ICE Speed

n_{sun}/n_s - Sun Shaft Speed

n_r - Ring Shaft Speed

p_{amb} - Ambient Pressure

T_{amb} - Ambient Temperature

v_{car} - Vehicle Speed

x_{pred} - Output Value of a NN

x_{real} - Output Value of a simulation

z_r - Ring Gear Number of Teeth

z_s - Sun Gear Number of Teeth

τ - Torque

τ_c - Torque at Carrier shaft

τ_{EM1}/τ_{EM2} - Torque from EM1 / EM2

τ_r - Torque at Ring Shaft

τ_s - Torque at Sun Shaft

J_i - Mass Moment of Inertia

$\dot{\omega}_i$ - Rotational Acceleration

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Appendix

FIGURE 19 Full-length single NN_{ICE} predictions compared to real values

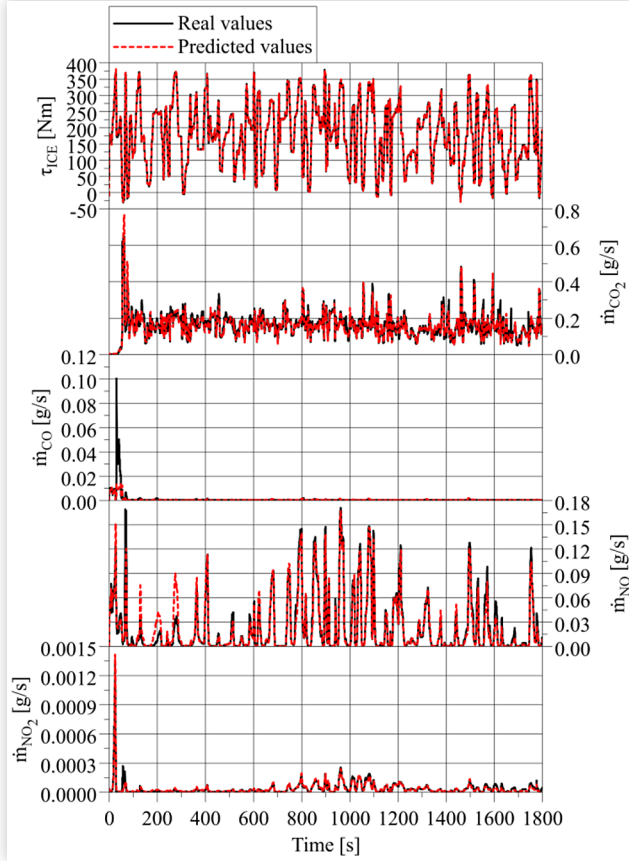


FIGURE 20 Full-length single $NN_{PG-EM\ v1-shift}$ prediction compared to real values

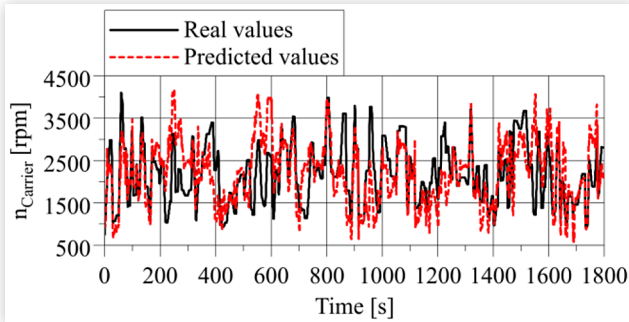


FIGURE 21 Model 1 full-length cascaded NN_{ICE} predictions compared to real values

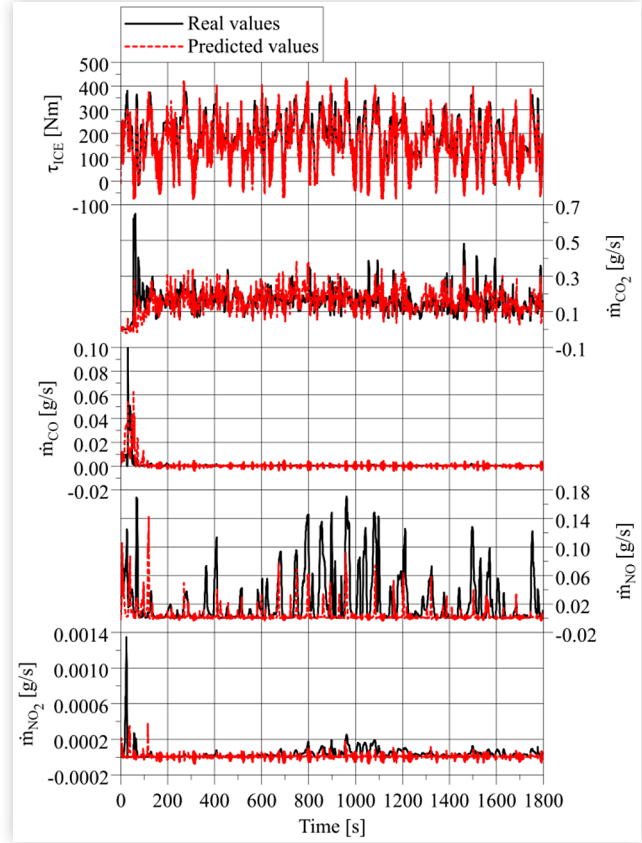


FIGURE 22 Model 1 full-length $NN_{PG-EM\ v1}$ and $NN_{PG-EM\ v1-shift}$ cascaded predictions compared to real values

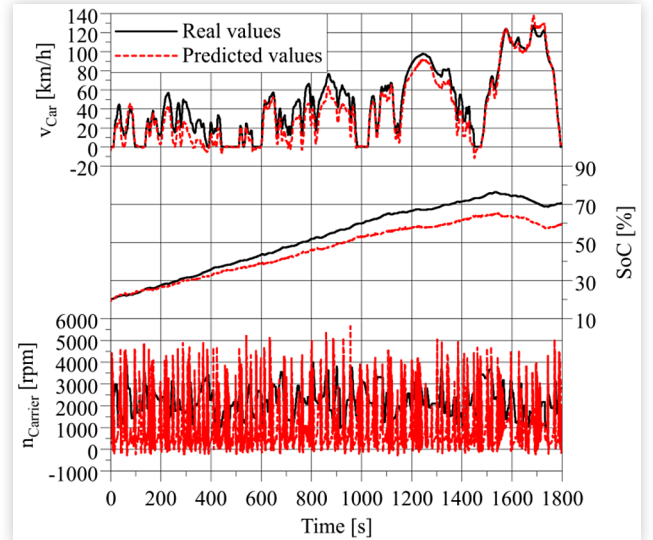


TABLE 10 Overview of error metrics comparison of Model 1 and Model 2 side by side

Output Parameter	Model 1		Model 2	
	MAE	MAPE	MAE	MAPE
τ_{ICE}	45.69 Nm	92.72 %	1.49 Nm	2.81 %
$\dot{m}_{CO_2}$	0.051 g/s	133,347.91 %	0.015 g/s	13,874.52 %
$\dot{m}_{CO}$	0.0016 g/s	10,636,515.64 %	0.00045 g/s	364,325.16 %
$\dot{m}_{NO}$	0.022 g/s	1,949.98 %	0.0042 g/s	137.04 %
$\dot{m}_{NO_2}$	4.10e-5 g/s	2,288.55 %	8.30e-6 g/s	159.09 %
V_{Car}	9.13 km/h	30,545.48 %	2.02 km/h	37,702.02 %
SoC	6.35 %	10.71 %	0.33 %	0.93 %